Cricket Match Analysis using SQL

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1. Executive Summary

The Cricket Match Analysis project was developed to analyze cricket match data using SQL and generate meaningful insights from historical match records. The project focuses on examining team performance, player achievements, venue statistics, toss decisions, and scoring patterns. By applying SQL queries on the cricket dataset, valuable business insights were extracted that help understand trends in match outcomes and team strategies.

The analysis demonstrates how SQL can be effectively used for data cleaning, data retrieval, aggregation, and business intelligence. This project showcases practical database management skills along with analytical thinking required for real-world data analysis.

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2. Business Problem

Cricket generates a large amount of match-related data every season. Without proper analysis, it is difficult to identify performance trends, compare teams, evaluate player contributions, or understand how factors such as toss decisions and venues influence match results.

The primary business problem addressed in this project is to transform raw cricket match data into actionable insights using SQL queries. These insights can help coaches, analysts, and management make informed decisions based on historical performance.

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3. Dataset Overview

The dataset contains historical cricket match records, with each row representing one completed match. It includes information related to teams, venues, scores, toss results, match winners, and player performances.

Main Attributes

Column	Description
Season	Tournament season
Home Team	Home team name
Away Team	Away team name
Toss Winner	Team winning the toss
Toss Decision	Bat or Field
First Innings Score	First innings total
Second Innings Score	Second innings total
Winner	Match winner
Player of the Match	Best player
Venue	Match venue
Start Date	Match date

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4. SQL Concepts Used

The project applies several SQL concepts to retrieve and analyze the data efficiently.

- SELECT Statement
- WHERE Clause
- ORDER BY
- GROUP BY
- Aggregate Functions (COUNT, SUM, AVG, MAX, MIN)
- DISTINCT
- LIMIT
- CASE Statement (if used)
- Date Functions
- String Functions
- Data Cleaning
- Sorting and Filtering

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5. Business Questions and Analysis

This section contains all SQL queries and their respective outputs.

Question 1

How many matches are available in the dataset?

SQL Query

```sql
-- Q1: Total matches played
SELECT
    COUNT(*) AS Total_Match
FROM
    cricket_data;
```

Output

The dataset contains **1017** matches, providing a sufficient sample size for performance and trend analysis.

Question 2

How many matches were played in each season?

SQL Query

```sql
-- Q2: Total matches played per season
SELECT
    season, COUNT(*) AS season_matches
FROM
    cricket_data
GROUP BY season;
```

Output

season	season_matches
2023	69
2022	74
2021	60
2020	60
2019	60
2018	60
2017	59
2016	60
2015	58
2014	60
2013	76
2012	74
2011	72
2010	60
2009	57
2008	58

Question 3

Which team has won the highest number of matches?

SQL Query

```sql
-- Q3: Most winning team
SELECT
    winner, COUNT(*) AS wins
FROM
    cricket_data
GROUP BY winner
ORDER BY wins DESC
LIMIT 1;
```

Output

MI emerged as the most successful team with **139** victories, demonstrating consistent performance throughout the tournament.

Question 4

Which are the top 5 most successful teams based on total match wins?

SQL Query

```sql
-- Q4: Top 5 winning teams
SELECT
    winner, COUNT(*) AS wins
FROM
    cricket_data
GROUP BY winner
ORDER BY wins DESC
LIMIT 5;
```

Output

winner	wins
MI	139
CSK	130
KKR	121
RCB	116
SRH	111

Question 5

How many matches were won by the team that also won the toss?

SQL Query

```sql
-- Q5: Team who wins both toss and match
SELECT
    COUNT(*) AS toss_matches_win
FROM
    cricket_data
WHERE
    toss_won = winner;
```

Output

Teams won both the toss and the match **521** times, suggesting that winning the toss provides a competitive advantage in many matches.

Question 6

What is the distribution of toss decisions (Batting vs Fielding) made by teams after winning the toss?

SQL Query

```sql
-- Q6: batting vs bowling decision analysis who wins toss
SELECT
    decision, COUNT(*) AS total
FROM
    cricket_data
GROUP BY decision;
```

Output

decision	total
BOWL FIRST	647
BAT FIRST	370

Question 7

What is the highest first innings score recorded in the dataset?

SQL Query

```sql
-- Q7: Highest score (1st inning)
SELECT
    first_inning_score AS high_score
FROM
    cricket_data
ORDER BY high_score DESC
LIMIT 1;
```

Output

The highest first-innings score of **263** reflects the batting potential under favorable pitch and weather conditions.

Question 8

What is the lowest first innings score recorded in the dataset?

SQL Query

```sql
-- Q8: Lowest score (1st inning)
SELECT
    MIN(first_inning_score) AS high_score
FROM
    cricket_data;
```

Output

The lowest first-innings score of **62** indicates exceptional bowling performances or challenging batting conditions.

Question 9

What is the highest successful run chase (second innings score) recorded in the dataset?

SQL Query

```sql
-- Q9: Highest run chase (2nd inning)
SELECT
    MAX(second_inning_score) AS high_chase
FROM
    cricket_data;
```

Output

A second-innings score of **226** represents one of the most successful run chases, showcasing the team's ability to perform under pressure.

Question 10

Which player has received the highest number of Player of the Match awards?

SQL Query

```sql
-- Q10: Maximum time Player of the match
SELECT
    pom, COUNT(*) AS Match_player
FROM
    cricket_data
GROUP BY pom
ORDER BY Match_player DESC
LIMIT 1;
```

Output

AB de Villiers received the highest number of Player of the Match awards, highlighting outstanding individual contributions.

Question 11

Who are the top 5 players with the most Player of the Match awards?

SQL Query

```sql
-- Q11: Top 5 Player of the Match
SELECT
    pom, COUNT(*) AS Match_player
FROM
    cricket_data
GROUP BY pom
ORDER BY Match_player DESC
LIMIT 5;
```

Output

pom	Match_player
AB de Villiers	25
Chris Gayle	22
Rohit Sharma	19
David Warner	18
MS Dhoni	17

Question 12

How many matches has each team played as the home team?

SQL Query

```sql
-- Q12: Home team performance
SELECT
    home_team, COUNT(*) AS home_matches
FROM
    cricket_data
GROUP BY home_team;
```

Output

home_team	home_matches
RCB	121
MI	119
KKR	113
DC	121
PBKS	21
SRH	122
LSG	14
GT	16
CSK	122
RR	96
KXIP	93
RPS	14
GL	15
PWI	23
Kochi	7

Question 13

How many matches has each team played as the away team?

SQL Query

```sql
-- Q13: Away team performance
SELECT
    away_team, COUNT(*) AS home_matches
FROM
    cricket_data
GROUP BY away_team;
```

Output

away_team	home_matches
GT	14
SRH	119
LSG	14
CSK	99
RR	109
RCB	118
DC	116
MI	126
KKR	124
PBKS	21
KXIP	97
RPS	16
GL	15
PWI	22
Kochi	7

Question 14

Which venue has hosted the highest number of cricket matches?

SQL Query

```sql
-- Q14: Venue who hosts maximum matches
SELECT
    venue_name, COUNT(*) AS host_venue
FROM
    cricket_data
GROUP BY venue_name
ORDER BY host_venue DESC
LIMIT 1;
```

Output

Wankhede Stadium, Mumbai hosted the highest number of matches, making it one of the tournament's primary venues.

Question 15

How many matches were won while successfully chasing the target?

SQL Query

```sql
-- Q15: Matches where team won by chasing
SELECT
    COUNT(*) AS chasing_win
FROM
    cricket_data
WHERE
    second_inning_score > first_inning_score;
```

Output

535 matches were won while chasing, indicating that batting second was often a successful strategy.

Question 16

How many matches were won by successfully defending the first innings total?

SQL Query

```sql
-- Q16: Matches where team defended score
SELECT
    COUNT(*) AS defend_score
FROM
    cricket_data
WHERE
    second_inning_score < first_inning_score;
```

Output

468 matches were successfully defended, demonstrating that strong bowling attacks and disciplined fielding remain crucial for victory.

Question 17

What are the average first innings and second innings scores across all matches?

SQL Query

```
-- Q17: Avg score per match
SELECT
    AVG(first_inning_score) AS avg_1st,
    AVG(second_inning_score) AS avg_2nd
FROM
    cricket_data;
```

Output

avg_1st	avg_2nd
163.8456	150.5270

Question 18

Which matches were closely contested, with a score difference of less than 10 runs?

SQL Query

```
-- Q18: Close matches(difference < 10 runs)
SELECT
    season, short_name
FROM
    cricket_data
WHERE
    (first_inning_score - second_inning_score) < 10
limit 10;
```

Output

season	short_name
2023	RCB v GT
2023	MI v SRH
2023	KKR v LSG
2023	PBKS v RR
2023	SRH v RCB
2023	LSG v MI
2023	CSK v KKR
2023	SRH v LSG
2023	KKR v RR
2023	MI v RCB

Question 19

What is the win percentage of each team based on the total number of matches in the dataset?

SQL Query

```sql
-- Q19: Team-wise win percentage
SELECT
    winner,
    COUNT(*) * 100 / (SELECT
        COUNT(*)
    FROM
        cricket_data) AS win_percentage
FROM
    cricket_data
GROUP BY winner
ORDER BY win_percentage DESC;
```

Output

winner	win_percentage
MI	13.6676
CSK	12.7827
KKR	11.8977
RCB	11.4061
SRH	10.9145
DC	10.4228
RR	10.0295
KXIP	8.6529
GT	2.1632
PBKS	1.8682
LSG	1.6716
RPS	1.4749
GL	1.2783
PWI	1.1799
Kochi	0.5900

6. Key Insights

Based on the SQL analysis, the following key insights were identified:

- Certain teams consistently performed better across multiple seasons.
- Winning the toss increased the probability of winning the match in many cases.
- Teams preferred fielding after winning the toss more frequently than batting.
- Some venues hosted significantly more matches than others.
- A few players dominated the Player of the Match awards.
- Average first innings scores indicated the general scoring trend.
- Teams chasing targets won a considerable number of matches.
- Team performance varied across seasons.

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7. Recommendations

Based on the analysis, the following recommendations can be made:

- Teams should analyze venue-specific performance before selecting playing strategies.
- Toss decisions should be based on historical venue statistics rather than fixed strategies.
- Consistent top-performing players should be retained as key members of the squad.
- Teams should prepare separate strategies for batting first and chasing targets.
- Performance trends should be reviewed every season to improve team planning.
- Historical data analysis can support better decision-making during tournaments.

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8. Conclusion

This project successfully demonstrates how SQL can be used to perform business analysis on cricket match data. By applying SQL queries, meaningful insights were extracted regarding team performance, player achievements, venue statistics, and scoring trends.

The project highlights the importance of SQL in data analytics and business intelligence. It also strengthens practical knowledge of database management, query writing, data aggregation, and analytical problem-solving.

Future improvements may include integrating the database with Power BI or Tableau to create interactive dashboards and visual reports.

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9. Skills Demonstrated

Throughout this project, the following technical and analytical skills were demonstrated:

Technical Skills

- SQL Query Writing
- Database Creation
- Data Import
- Data Cleaning
- Aggregate Functions
- GROUP BY Analysis
- Filtering and Sorting
- Business Data Analysis

Analytical Skills

- Problem Solving
- Data Interpretation
- Business Insight Generation
- Performance Analysis
- Decision Support

Tools Used

- MySQL
- VS Code / MySQL Workbench